A Firm-Level Microsimulation for VAT Policy Analysis

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Motivation

How should the UK cost reforms to its VAT registration threshold — changes to its level, shape, or rate — on a single, common, reproducible basis,

when the firm-level data needed are not openly available?

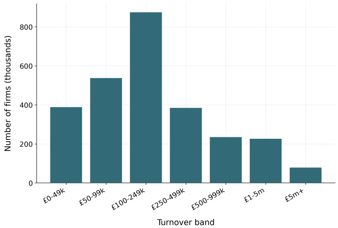
Why the VAT threshold matters

- The UK VAT threshold is a notch, not a kink: cross it and 20% falls due on the firm's *entire* turnover — not just the slice above.
- A discrete liability jump of $\tau T^* = £17{,}000$ ($= 20\% \times £85{,}000$) at the threshold.
- Threshold was £85,000 (2023–24), raised to £90,000 on 1 April 2024 — after a seven-year nominal freeze, the longest in the tax's history.
- Among the highest thresholds in the OECD, and it sits in a *dense* part of the firm-size distribution.
- Firms respond by bunching just below it; the cliff-edge keeps the threshold under recurrent pressure for reform.

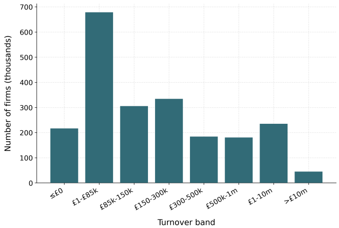
▸ notch vs. kink

Firms cluster just below the threshold

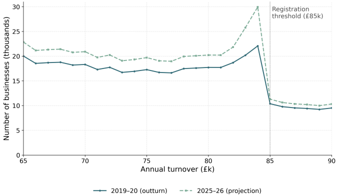
Cross £85,000 and 20% VAT falls due on the firm's *entire* turnover — a £17,000 jump — so firms cluster right below it.



All UK firms by turnover band (ONS)



VAT-registered firms by band (HMRC)



Bunching below £85k (OBR, 2023)

This paper

- An open, reproducible firm-level microsimulation that costs UK VAT threshold reforms on a common static basis.
- Built on synthetic firm microdata calibrated to published HMRC + ONS aggregates — released openly and regenerable under any counterfactual schedule.
- Prices three reform families — level (move the threshold), shape (a graduated taper), rate (a reduced-rate band) — three ways:
  1. **statically** (mechanical reclassification);
  2. by the exact, elasticity-free **dominated-region** distortion each removes;
  3. **behaviourally**, conditional on an assumed turnover elasticity e.
- Headline: only the graduated taper removes the notch's distortion; raising the threshold is the dearest, least efficient option.

Related literature

- **Bunching / notch methodology** — source of the "dominated region"
  - Saez (2010); Kleven and Waseem (2013); Kleven (2016)
- **VAT registration threshold (empirical)**
  - Liu et al. (2021) (UK notch, ∼43% voluntary registration); Liu, Lockwood and Tam (2024); Onji (2009); Asatryan and Peichl (2017); Waseem (2024)
- **Does bunching identify a structural elasticity?**
  - Blomquist et al. (2021); Bertanha et al. (2023) — *not* non-parametrically identified
- **Structural models of size-based thresholds**
  - Garicano et al. (2016); Gourio and Roys (2014)

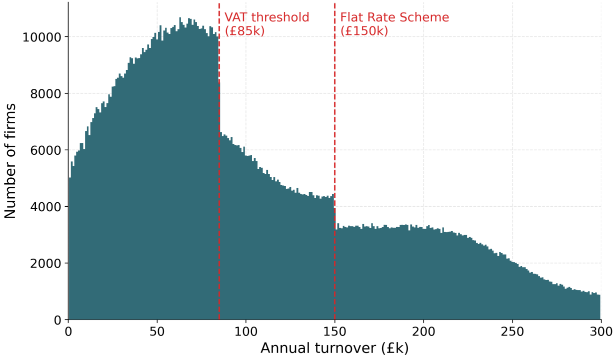
Data and calibration

Data: synthetic firms calibrated to public aggregates

- **No open firm-level VAT microdata exist** $\Rightarrow$ build a synthetic population calibrated to published aggregates (data year 2023–24).
- **Calibration targets:**
  - ONS *UK Business: Activity, Size and Location* — firm counts by sector × turnover band
  - HMRC *VAT Annual Statistics* — registered counts and net VAT liability by band / sector
- **Scale:** ≈2.94m firm records, weighted to represent ≈2.5m UK firms.
- **Accuracy:** ≈90% across five calibrated dimensions.
- Approach mirrors open household tax-benefit microsimulation (Ghenis et al., 2022).

▸ how firms are built ▸ distribution

The calibrated turnover distribution



Density steps down at the £85,000 registration threshold and again at the £150,000 Flat Rate Scheme ceiling. The step is inherited from coarse HMRC band targets — quantified later by a placebo test.

Static costing

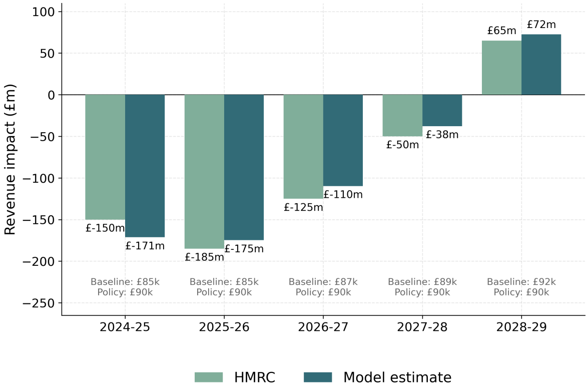
Static costing: mechanical reclassification

Static revenue at threshold T

$$R(T) = \sum_i w_i\, v_i\, \mathbf{1}\{y_i \geq T\}.$$

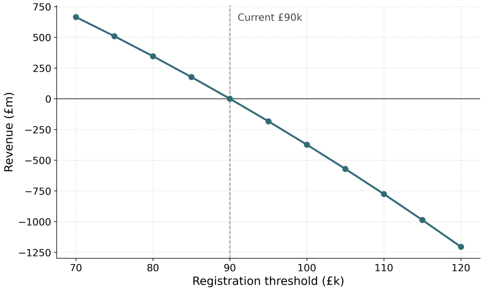
- Turnover held fixed $\Rightarrow$ revenue moves only through firms whose registration status flips (those between the old and new cutoff).
- Three reform levers, four schedules, all on one base:
  - **Level** — move the threshold up or down
  - **Shape** — graduated taper, 0% at £85k $\rightarrow$ 20% at £105k
  - **Rate** — reduced-rate band (10% or 15%) in £85k–£105k
- For each, recompute $R = \sum_i w_i v_i$ under the new schedule and difference.

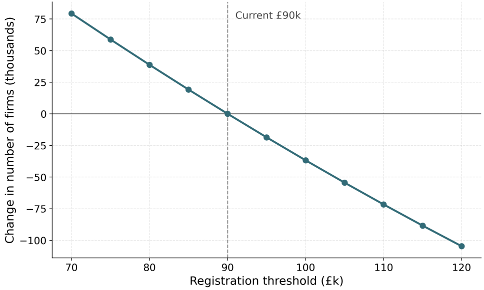
Validation: the April-2024 anchor reform



- Cost the £85k→£90k April-2024 rise, my estimate vs HMRC's published costing.
- 2025–26: £175m vs £185m — agree within £7–21m per year across the horizon.
- Both turn to a small *gain* by 2028–29 as fiscal drag lifts the frozen baseline.
- Microdata are calibrated to HMRC aggregates ⇒ a consistency check, not out-of-sample validation.

Static sweep: cost of moving the threshold (2025–26)





From a £90k base (£203.8bn), the schedule is roughly linear, ≈£150–220m per £5k step: e.g. −£374m to raise to £100k, +£665m to lower to £70k. Even a £30k rise to £120k costs ∼£1.2bn — under 0.6% of the base.

The reform menu, one common base

Reform	Lever	Static	Dominated region
Raise to £100k	Level	−£508m	relocated
Taper £85–105k	Shape	−£336m	removed
Reduced 10%	Rate	−£343m	split in two
Reduced 15%	Rate	−£171m	split in two

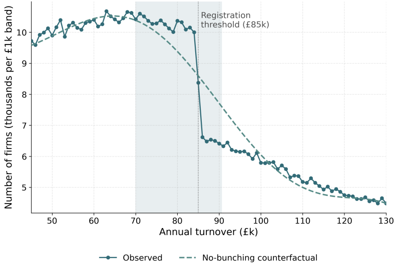
Common £85,000-notch / £183.6bn 2023–24 base. The level move is the most expensive yet removes no distortion; the taper removes it entirely for less.

"Split in two" = a second notch appears where the reduced-rate band ends (£105k reverts to 20%), so the empty band is fragmented, not shrunk.

▸ behavioural

The notch’s distortion

Bunching in the synthetic data is mechanical

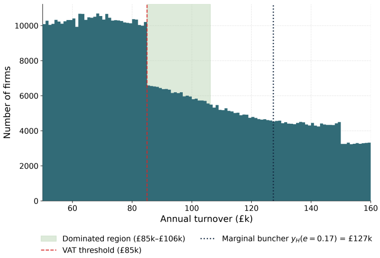


Excess mass over a polynomial no-bunching counterfactual g:

$$b = \frac{B(y^*)}{g(y^*)} \approx \frac{\Delta y^*}{y^*}, \qquad E = \sum_{\text{window}} (n_j - c_j).$$

- Reproduced density step: ratio $b = 0.060$ (SE 0.005), excess mass $E = 8{,}712$ firms (SE 662).
- Placebo: remove the step from the calibration target $\Rightarrow$ E collapses to 0–99 firms.
- So the step is inherited from coarse HMRC bands, not firm behaviour — the generator has no location choice.
- Genuine threshold bunching remains the administrative fact of Liu et al. (2021).

The dominated region: an exact, elasticity-free distortion



The notch in the firm's profit:

$$\pi(y) = \begin{cases} y - c(y), & y < T^*, \\ (1-\tau)\,y - c(y), & y \geq T^*. \end{cases}$$

- At T^*, profit jumps *down* by $\tau T^* = £17{,}000$ — VAT on turnover already earned.
- The dominated region is the band just above T^* where no firm locates (strictly worse than at T^*):

$$a = T^* \frac{\tau}{1-\tau} = £21{,}250.$$

- Interval (£85,000, £106,250), ≈137,000 weighted firms.
- Depends on τ and T^* alone — no elasticity, no cost specification. An order of magnitude wider than a kink's ∼£1k.

Which reforms actually shrink the band?

- **Raise the threshold** — relocates the £21,250 band unchanged to the new cutoff.
- **Banded reduced rate** — does not shrink it. Lowers the single notch (£17,000 $\to$ £12,750 at 15%) but adds a secondary notch where the band reverts at £105k:

$$a' = T_1 \frac{\tau - r}{1 - \tau}; \qquad a + a' = \text{£}21{,}562\ (15\%), \quad \text{£}22{,}569\ (10\%).$$

  It splits one wide empty band into two, not shrinks it.
- **Graduated taper** — continuous, no notch and no dominated region: it removes the distortion entirely.

Only the taper removes the distortion — and for less (£336m).

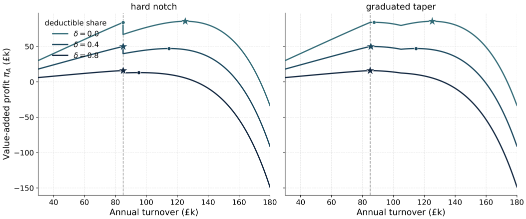
Behavioural layer

The firm's problem: VAT on *value added* (formulation A)

A firm of ability n chooses turnover y to maximise

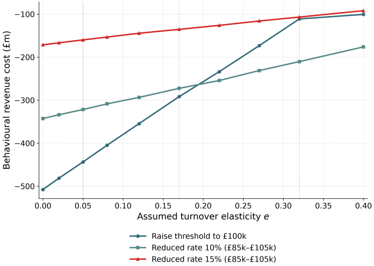
$$\pi(y) = (1-\delta)\big(1-\tau f(y)\big)\, y - C(y; n, e).$$

- VAT falls on value added $(1-\delta)y$, not whole turnover; δ is the firm's deductible-input share.
- C is the firm's own-factor cost (labour, capital): inputs δy are already netted by $(1-\delta)$, so reading C as total cost would double-count them.
- Optimum $y^\star = n\big[(1-\delta)(1-\tau)\big]^e$. δ is parametric for now — ONS Supply–Use value-added shares to come.



Higher δ scales the curve down and pulls the registered optimum toward the threshold (eventually tipping to bunching).

Behavioural layer: conditional on an assumed elasticity



Each firm rescales turnover under the reform:

$$y_i^\star = y_i\left(\frac{1-\tau_{\text{reform}}}{1-\tau_{\text{base}}}\right)^e, \qquad \frac{\partial \ln y}{\partial \ln(1-\tau)} = e.$$

- Iso-elastic simulator (value-added base, formulation A) re-optimises each firm (left); the elasticity e is assumed, not identified.
- Sweep $e \in \{0.05, 0.17, 0.32\}$. At $e = 0.17$ base-broadening offsets much of each static loss:
  - raise to £100k: £508m $\to$ £292m
  - 10% band: £343m $\to$ £273m
  - 15% band: £171m $\to$ £135m
- Static cost is the $e \to 0$ limit; only $e = 0.05$ has an external anchor (Kleven and Waseem, 2013).
- Taper excluded: its marginal-rate channel is not credibly identified iso-elastically.

Conclusion

Conclusion

- What this is: an open, reproducible firm-level microsimulation that models the UK VAT threshold as a turnover-tax notch on synthetic firm data, and prices level / shape / rate reforms three ways.
- Static: reproduces HMRC's anchor costing within $\sim$£10m; raising to £100k costs £508m, the taper £336m.
- Distortion: the dominated region $a =$ £21,250 is exact and elasticity-free — only the taper removes it; the level move merely relocates it; a reduced rate splits it in two.
- Transparency: synthetic bunching is mechanical (placebo: 8,712 $\rightarrow\sim$ 99 firms) — no elasticity is read off the data.
- Behavioural offsets are conditional on assumed e; the static cost and dominated region are the elasticity-free anchors.
- Next: identify e from administrative microdata; add welfare / MVPF accounting; sectoral incidence.

Thank you!

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Scan: paper, data & code

Appendix: how the synthetic firms are built

- Draw firms by sector × ONS turnover band; turnover smoothed within band:

$$y_i = \underline{b} + (\overline{b} - \underline{b})\, u_i + \varepsilon_i, \quad u_i \sim \mathcal{U}(0,1),\ \varepsilon_i \sim \mathcal{N}(0, \sigma_b^2).$$

- Input share $x_i = \rho_i y_i$ ($\rho_i \in [0.6, 1.5]$, rescaled Beta) $\Rightarrow$ net VAT base $v_i = y_i - x_i$.
- Each firm gets a strictly positive weight $w_i = e^{\theta_i}$; weighted target $\widehat{T}_k(\theta) = \sum_i a_{ki} w_i$.
- Fit θ by gradient descent on a symmetric relative-error loss:

$$L(\theta) = \frac{1}{K} \sum_k \lambda_k \min\{(\widehat{T}_k / T_k - 1)^2,\ (T_k / \widehat{T}_k - 1)^2\} + \frac{\alpha}{N} \sum_i |\theta_i|.$$

- The threshold is a generator parameter: every counterfactual — higher cutoff, taper, reduced rate — regenerates a fresh, internally consistent population.

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Appendix: notch vs. kink

- A kink taxes only the turnover *above* the threshold — the marginal rate jumps, but liability is continuous.
- A notch taxes the firm's *entire* turnover once it registers — liability jumps discretely by $\tau T^* = £17{,}000$.
- The UK VAT threshold is a notch $\Rightarrow$ a strict dominated region exists (no firm should locate just above T^*), whereas a kink leaves only a $\sim$£1k smoothed gap.
- This is why the distortion can be measured exactly and without any elasticity.

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Appendix: behavioural costs across e

Reform	Static (e=0)	e=0.05	e=0.17	e=0.32
Raise to £100k	−£508m	−£444m	−£292m	−£111m
Reduced 10% band	−£343m	−£322m	−£273m	−£210m
Reduced 15% band	−£171m	−£160m	−£135m	−£107m

A larger e makes every flat reform cheaper (base-broadening). The graduated taper is omitted — its base-broadening and marginal-rate channels cannot both be netted in the iso-elastic cost. The $e \to 0$ limit reproduces the static costs to within ∼£1m.

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Appendix: secondary-notch arithmetic (reduced rate)

A reduced rate r in $[T^*, T_1]$ with $T_1 = £105{,}000$:

- **Primary** notch shrinks: jump rT^*, width $a = T^* \dfrac{r}{1-r}$ = £15,000 (15%), £9,444 (10%).
- **Secondary** notch at the band top, where rate reverts to τ: jump $(\tau - r)T_1 = £5{,}250$ (15%), width $a' = T_1 \dfrac{\tau - r}{1-\tau}$ = £6,562 (15%), £13,125 (10%).
- **Total** dominated width $a + a' = £21{,}562$ (15%), £22,569 (10%) — both marginally *above* the £21,250 baseline.

The reduced rate splits the empty band; it does not shrink it.

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Appendix: limitations and future work

- **Value added, parametric δ** — the behavioural simulator now taxes *value added* $(1-\delta)y$ (formulation A), but δ is parametric, not yet measured firm by firm (ONS Supply–Use to come); the dominated region still uses the statutory-rate notch, and $\sim$43% of below-threshold firms register voluntarily (Liu et al., 2021), which the model abstracts from.
- **Partial efficiency metric** — dominated-region width indexes misallocation only, omitting compliance-cost savings (Keen and Mintz, 2004); a full welfare / MVPF accounting is needed.
- **Aggregate only** — no sectoral incidence (VAT liability by sector excluded from the optimiser).
- **Synthetic, partial-equilibrium** — abstracts from price pass-through; no genuine firm location choice.